RL Lecture 2 Worksheet

Markov processes, returns, Bellman equations and policies

Name: _____ ID: _____ Date: _____

Core practice: 60 marks | Approximately 50-60 minutes | Calculator allowed.
Optional extension: value iteration, 10 additional marks.

Conventions: R(s) is received on the next transition **from** state s. In an MDP, R(s,a) is received when action a is taken in s. Terminal states have zero future value. Round final values to three decimal places.

1 Markov property and transitions [12]

(a) Decide whether each proposed state is Markov **under the stated model**. Give one short reason. [2 each]

Proposed state and model	Markov? Why?
Current weather; tomorrow depends only on today's weather.	
Current room temperature; the next temperature also depends on whether the heater is on, which is omitted.	
Robot position and battery level; the next-state distribution depends only on these and the chosen action.	

(b) Rows are current states and columns are next states. Complete the two missing entries so that each row is a probability distribution. [2]

From / To	C1	C2	Sleep
C1	0.50	_____	0
C2	0	0.75	_____
Sleep	0	0	1

(c) Starting in C1, what is the probability of being in Sleep after exactly two transitions? Show the contributing path. [2]

_____

_____

(d) What does the Sleep row mean, and why can an episode still stop on arrival there? [2]

_____

_____

2 Rewards, returns and values [12]

Use this single sampled episode from the lecture's student MRP:

C1 → C2 → C3 → Pass → Sleep

State being left	C1	C2	C3	Pass	Sleep
Reward R(s)	−2	−2	−2	+10	0

The episode ends on entering Sleep. For this exercise, the listed reward is deterministic on departure from each state.

(a) Write the four rewards received along the episode. Use them to compute the return starting in C1 for each discount factor. [6: 2 per row]

Reward sequence: (_____ , _____ , _____ , _____)

$G = R_1 + \gamma R_2 + \gamma^2 R_3 + \gamma^3 R_4$

Discount γ	Substitution / working	Return G
0		
0.5		
1		

(b) Is +10 received on the C3 → Pass transition or on Pass → Sleep? Explain using the reward convention. [2]

_____

_____

(c) Four independent episodes starting in C1 give returns −2.250, −3.125, −3.500 and −3.125, using the same discount. Estimate v(C1) by averaging these four returns. Count only the initial C1 in each episode. [2]

_____

_____

(d) Explain why one sampled return need not equal the state's true value. [2]

_____

_____

3 Solving a Bellman equation [16]

Consider an MRP with states A, B and terminal T. Let $\gamma = 0.5$.

Current state	Immediate reward	Next-state probabilities
A	−1	B with 0.5; T with 0.5
B	+2	B with 0.5; T with 0.5
T	0	T with 1.0

Rewards depend on the state being left. Use $v(T) = 0$. The self-loop at B means its reward may be earned more than once.

(a) Complete P in the order [A, B, T], and write R as a column vector. [4: P rows 1 each; R 1]

From / To	A	B	T
A			
B			
T			

R = (_____ , _____ , _____)^T

(b) Write Bellman equations for A and B, substituting γ and the probabilities. [4]

v(A) = _____

v(B) = _____

(c) Solve these equations. Start with v(B), then use it to find v(A). [4]

_____

_____

_____

(d) If $\gamma = 0$, what are v(A) and v(B)? [2]

v(A) = _____ v(B) = _____

(e) Someone proposes $v(A) = 0$, $v(B) = 4$, $v(T) = 0$ for $\gamma = 0.5$. Check the Bellman equation for B to decide whether this is a valid value function. [2]

_____

_____

4 Decisions, policies and optimal values [20]

An MDP has decision state S, intermediate state U, and terminal T. Let γ = 0.9. These are all available actions:

State	Action	Immediate reward	Next state
S	Safe	+2	T with probability 1
S	Invest	−1	U with probability 1
S	Chance	+1	U with 0.5; T with 0.5
U	Finish	+6	T with probability 1

T ends the episode and v(T) = 0. The fixed policy at S chooses Safe with probability 0.50, Invest with 0.25, and Chance with 0.25. At U, it always chooses Finish.

(a) What does π(Invest | S) = 0.25 mean? [1]

_____

(b) Find vπ(U), then compute qπ(S,a) for each action. [8: vπ(U) 2; each q-value 2]

vπ(U) = _____

Action a	Reward + discounted expected next-state value	qπ(S,a)
Safe		
Invest		
Chance		

(c) Use the policy probabilities to calculate vπ(S). [3]

_____

_____

(d) Identify an optimal action at S and give v*(S) and v*(U). [6: 2 each]

Action: _____ v*(S): _____ v*(U): _____

Working: _____

(e) Explain the difference between evaluating a fixed policy and finding an optimal policy. [2]

_____

_____

5 Optional bridge to value iteration [10]

Use the **same MDP as Question 4**, with $\gamma = 0.9$. This extension connects the lecture's optimal-value idea to the interactive animation.

Initialize: $V_0(S) = V_0(U) = V_0(T) = 0$. Keep $V_k(T) = 0$.
Synchronous updates: all values in row k + 1 use values from row k.

$V_{k+1}(s) = \max_a [R(s,a) + \gamma \Sigma_{s'} P(s' \mid s,a) V_k(s')]$

(a) Complete three sweeps. [6: 1 per blank]

Sweep k	$V_k(S)$	$V_k(U)$	$V_k(T)$
0	0	0	0
1			0
2			0
3			0

(b) For the first update of S, list the three candidate action values. Which action is greedy at this stage? [2]

Safe: _____ Invest: _____ Chance: _____ Greedy: _____

Now list the candidates for the second update of S.

Safe: _____ Invest: _____ Chance: _____ Greedy: _____

(c) Why does S not receive the full benefit of U's reward during the first synchronous sweep? [1]

_____

_____

(d) What replaces the maximum if we instead evaluate the fixed policy in Question 4? [1]

_____

_____

Class discussion

- Why can the greedy action change as value estimates improve?
- Does a stochastic transition mean the agent chooses the successor state?
- Why is the value of T zero even though the transition into it can deliver a reward?

Source: adapted from the uploaded David Silver Lecture 2 on Markov Decision Processes. Numerical exercises use simplified, self-contained examples.